Set partitions (Bell numbers) and involutions

Notes from Carlos' lessons

Source lessons: 07-01 (partitions of a set), 07-14 (number of partitions), 07-14 (number of involutions).

Both results use the EGF toolkit from the topic *Exponential generating functions*: the EGF of a product counts labelled pairs, e^x is "sets", and $e^{A(x)}$ is "a set of A-structures".

1 Set partitions: $\sum_n b_n \frac{x^n}{n!} = e^{e^x-1}$

b_n is the number of partitions of $[n] = \{1, \dots, n\}$ into non-empty blocks (the Bell numbers). Small cases from the 07-01 board: $b_1 = 1$ ($\{1\}$); $b_2 = 2$ ($\{1,2\}$; $\{1\}\{2\}$); $b_3 = 5$ ($\{1,2,3\}$; $\{1,2\}\{3\}$; $\{1\}\{2,3\}$; $\{1,3\}\{2\}$; $\{1\}\{2\}\{3\}$). (The board also draws the 3-vertex graph with all edges and with none, anticipating the interpretation below.)

Theorem 1. *$B(x) = \sum_{n\geq 0} b_n \frac{x^n}{n!} = e^{e^x-1}$, with $b_0 = 1$.*

Proof (07-14 board).

1. A **set** structure has exactly one structure on every label set: its sequence is $(1, 1, 1, \dots)$, with EGF $\sum_{n\geq 0} \frac{x^n}{n!} = e^x$. **Non-empty** sets exclude $n = 0$: the EGF is $e^x - 1$.
2. A partition into exactly n blocks is n non-empty sets with distinct label sets. $(e^x-1)^n$ counts them as an ordered n-tuple (the product rule, applied repeatedly). Since the label sets are disjoint and non-empty, the $n!$ orderings are all different, so $\frac{(e^x-1)^n}{n!}$ counts **unordered** collections of n non-empty blocks (the board: "n nonempty sets (unordered)").
3. Summing over the number of blocks,

$$B(x) = \sum_{n\geq 0} \frac{(e^x-1)^n}{n!} = e^{e^x-1}.$$

□

Equivalent graph picture (07-01). b_n is also the number of graphs on the vertex set $[n]$ whose connected components are cliques. The components of such a graph form a partition of $[n]$, and conversely a partition determines the graph. The board's triangle and three isolated vertices are the two extreme cases for $n = 3$.

Consequences (derived from the theorem)

- **Recurrence.** $B'(x) = e^x B(x)$ (chain rule), so comparing $[x^n]$ with the EGF product rule gives $b_{n+1} = \sum_{k=0}^{n} \binom{n}{k} b_k$. (Combinatorially: the block containing element $n+1$ has $n-k$ other elements, chosen in $\binom{n}{n-k}$ ways, and the other k elements are partitioned freely.) This gives $1, 1, 2, 5, 15, 52, 203, 877, \ldots$.
- **Dobiński's formula.** $e^{e^x - 1} = e^{-1} \sum_{k \geq 0} \frac{e^{kx}}{k!}$, and $[x^n/n!]\, e^{kx} = k^n$, so $b_n = \frac{1}{e} \sum_{k \geq 0} \frac{k^n}{k!}$. This is probably what the $\sum \frac{k^n}{k!}$ scribbles on the 07-01 board are reaching for. The board writes $b_n = \sum_{k=1}^{n} \frac{k^n}{k!}$, which is not correct as written: it is missing the factor $1/e$, and the sum should run to ∞; for $n = 2$ it gives $1 + \frac{4}{2} = 3 \neq 2$.
- **The recurrence attempt on the board.** The board also tries $b_n = b_{n-1} + \sum_k k\, b(n-1, k)$, where $b(n, k)$ is the number of partitions of $[n]$ into k blocks: element n is either a new singleton block (b_{n-1} ways) or joins one of the k blocks of a partition of $[n-1]$ with k blocks ($k\, b(n-1, k)$ ways). That recurrence is correct. The next line on the board, "$b_n = b_{n-1} + b_{n-1}$" with a "??", looks like a stalled step: $\sum_k k\, b(n-1, k)$ is not equal to b_{n-1} in general (for $n = 3$: $b(2,1) \cdot 1 + b(2,2) \cdot 2 = 3 \neq b_2 = 2$), so that line should not be read as a valid identity.

2 Involutions: $\sum_n i_n \dfrac{x^n}{n!} = e^{x + x^2/2}$

A **permutation** of $[n]$ is a bijection $\pi : [n] \to [n]$; an **involution** is a permutation with $\pi \circ \pi = \mathrm{id}$ (for example the swap $1 \leftrightarrow 2$ on $\{1, 2\}$, or the identity). i_n is the number of involutions of $[n]$.

Structure. Draw the functional graph $x \to \pi(x)$. If $\pi \circ \pi = \mathrm{id}$, then for each x either $\pi(x) = x$ (a fixed point, a singleton), or $\pi(x) = y \neq x$ and $\pi(y) = \pi(\pi(x)) = x$ (the pair $\{x, y\}$ is swapped). Conversely, any partition of $[n]$ into singletons and pairs defines an involution (fix the singletons, swap the pairs). So

$$\text{involutions of } [n] \iff \text{sets of singletons and pairs partitioning } [n].$$

Theorem 2. $I(x) = \sum_{n \geq 0} i_n \frac{x^n}{n!} = \exp\!\big(x + \frac{x^2}{2}\big)$.

Proof. The block structure ("a single block, which is a singleton or a pair") has exactly one structure on a label set of size 1, one on a label set of size 2, and none otherwise. Its EGF is $\frac{x}{1!} + \frac{x^2}{2!} = x + \frac{x^2}{2}$. An involution is a *set* of such blocks, so by the $e^{A(x)}$ principle $I(x) = e^{x + x^2/2}$. □

Note on the board. It circles e^{x+x^2} and expands $\sum \frac{(x+x^2)^n}{n!}$; the pair term should be $\frac{x^2}{2!}$, since a pair $\{x, y\}$ is one unordered structure on 2 labels. Check with $n = 2$: $[x^2]\, e^{x+x^2/2} = \frac{1}{2} + \frac{1}{2} = 1$, so $i_2 = 2! \cdot 1 = 2$ (the identity and the swap). With x^2 in place of $\frac{x^2}{2}$ one would get $i_2 = 3$, which is wrong.

Consequences (derived)

- $I'(x) = (1+x)I(x)$, so $i_{n+1} = i_n + n\, i_{n-1}$ (element $n+1$ is fixed, or swapped with one of the n others).
- Explicit count (choose k disjoint pairs): $i_n = \displaystyle\sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^k k!(n-2k)!}$.
- Values: $1, 1, 2, 4, 10, 26, 76, 232, 764, 2620$ (checked with both formulas).

Related pattern

All three (sets, partitions, involutions) are instances of one picture: a structure on a label set has an EGF $A(x)$, and a **set** of such structures has EGF $e^{A(x)}$. Permutations as sets of cycles are the same with $A(x) = -\ln(1-x)$ (the cycle EGF has coefficients $\frac{(n-1)!\,x^n}{n!}$), giving $e^{-\ln(1-x)} = \frac{1}{1-x}$, i.e. $n!$ permutations. That is not on these boards, but it is a good sanity check of the principle.